TRIDO Motor Controller Manual

Version 2 (Last Updated 01/02/2023)

Applies to Version 01/20, 01/35, 01/50, 01/70 of the Software

TRIDO Motor Controller Manual 1
1. Introduction 3
1.1 This controller features: 3
1.2 Physical Requirements 3
1.3 Power Requirements 3
2. External Connectors: 3
Important Safety Considerations 3
2.1 Power Supply and Output 4
2.2 Controller Inputs and Device Control Outputs 5
2.2.1 Digital Outputs: 5
2.2.2 Digital Inputs: 5
2.2.3 Analog Inputs: 5
2.2.4 RS485 6
2.2.5 Side Access Connections: 6
2.3 Other features: 6
3.0 Controller Operation – Compressors 7
3.1 Features Keypad Operations: 7
3.2 Menu Page Description: 8
3.3 Error Message Description: 15
3.4 Register Map: 16
4.0 Controller Back Mounting Plate Drawing: 17

1. Introduction

1.1 This controller features:

- A wide range of operating voltages: from +18 to +32 VDC.
- An extended operating temperature range: from -40 °C to +40 °C.
- A user-friendly interface with a keypad and LCD for easy configuration and operation monitoring.
- An advanced motor control system, allowing motor speed from 0 to 350 RPM.
- Non-volatile memory for storage of configuration parameters and field-upgradable software.
- A non-volatile, resettable, accumulating revolutions counter
- External inputs and outputs allow communication and interfacing with external sensors and third-party control systems.
- An intelligent recovery mechanism for unattended, trouble-free operation.
- Support for remote control in the Modbus Slave mode and remote monitoring via Modbus protocol the all other modes.

1.2 Physical Requirements

The controller requires a minimum area of 6.25" wide and 7.25" to mount. Diagram 1

When mounting the controller, allow sufficient clearance around connectors to run wiring to minimize stress on the connectors and the circuit board.

1.3 Power Requirements

The controller can operate on a supply from +18 to +32 VDC. The power supply must be capable of supplying a sustained current of 12 Amps to ensure reliable operation and motor start-up.

Motor Model	Maximum Current
20	6 Amps
35	8 Amps
50	10 Amps
70	12 Amps

The controller has an internal fuse set at 10 Amp to protect from overcurrent. The internal fuse needs to be replaced if this fuse is blown.

2. External Connectors:

Important Safety Considerations



WARNING

EXPLOSION HAZARD - DO NOT REPLACE COMPONENTS UNLESS POWER HAS BEEN SWITCHED OFF OR THE AREA IS KNOWN TO BE NON-HAZARDOUS.

.

2.1 Power Supply and Output

The controller features several external connectors. These are color-coded to identify their functions:

- Black connectors are used to connect the power supply and motor.
- Green connectors are used for control, sensing, and remote communications. (not in service for the prototype testing.).

WARNING

INSTALL A CUTOFF SWITCH ON THE +24 VDC POWER RAIL BETWEEN THE CONTROLLER AND POWER SUPPLY. THIS ALLOWS FOR SAFE SERVICING OF THE CONTROLLER AND ELIMINATES ANY ARCING CAUSED BY MANUAL CONNECTION OR DISCONNECTION OF THE SUPPLY RAIL. FAILURE TO DO SO MAY RESULT IN PERSONAL INJURY AND OR DAMAGE TO THE CONTROLLER.



+24V 0V ⏚ W V U

Power Supply Connections Motor Connections

NOTES

Connect the ⏚ terminal to the local chassis or Earth ground, based on your installation.

As with most digital controllers, the Trido Energy Services Inc. controller Is sensitive to electronic noise at the installation site. Make sure all ground points in the location are connected using at least #6 wire.

The TRIDO motor is a 3 Phase, AC Permanent Magnet motor. Do not connect U, V, or W terminals to ground! Only connect to motor connection points only!

Reverse connection of any 2 of the motor wires to change the direction of the motor.

THE CONNECTION BETWEEN THE MOTOR AND THE CONTROLLER SHOULD BE CONTINUOUS. DO NOT INSTALL A SWITCH IN THIS CIRCUIT. INTERRUPTING THE CONNECTION BETWEEN THE MOTOR AND CONTROLLER WHILE THE ACTUATOR IS OPERATING CAN RESULT IN THE FAILURE OF THE CONTROLLER.

A SAFETY SWITCH CAN BE INSTALLED EITHER IN THE CONTROLLER SUPPLY LINE OR BY USING THE LOCK ON/LOCK OFF FEATURE OUTLINED IN THE SECTION 2.3.2 OF THIS MANUAL.

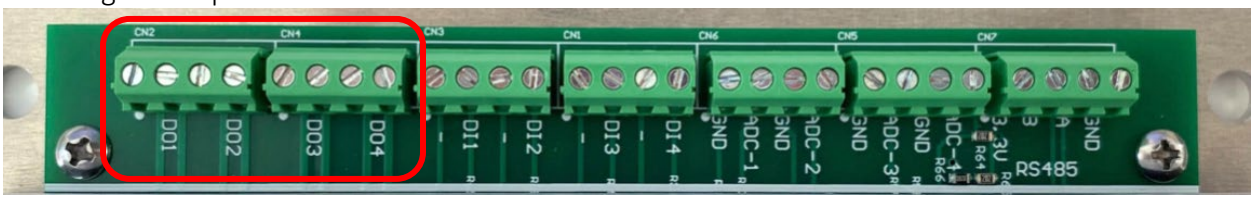
TRIDO recommends using #10 straight or locking spade lugs when hooking up to the black connectors.

The green connectors can readily accept 18 to 24 AWG solid or stranded wire.

2.2 Controller Inputs and Device Control Outputs

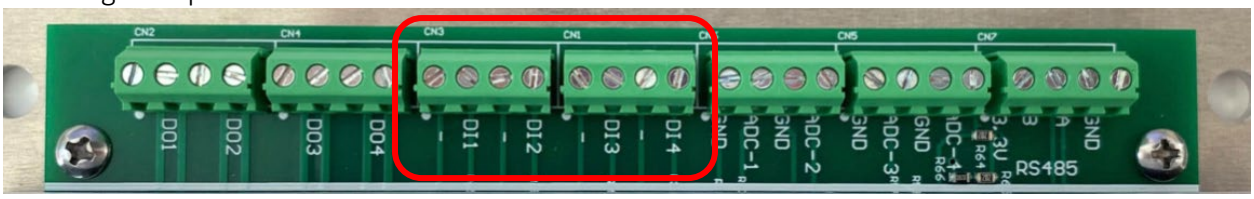
TRIDO controller has four digital inputs, four analog inputs, and four digital outputs.

2.2.1 Digital Outputs:



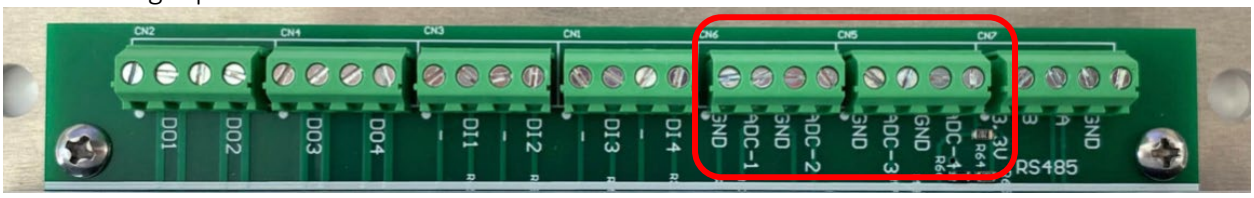
The illustration above shows the controller's location of 4 programmable digital outputs DO1, DO2, D03, and DO4. These four output sets can accept a 24 Volt signal that activates a 2 Amp solid state relay. Each signal can be separately controlled through conditional logic in application software developed for the function.

2.2.2 Digital Inputs:



Four optically separated digital inputs, DI1, DI2, DI3, and DI4, are provided to allow the input of digital singles for external devices. These singles are one of two situations, and the circuit is open or closed. These inputs can be used to convey signals from external devices that can then be used as part of controller programs to control.

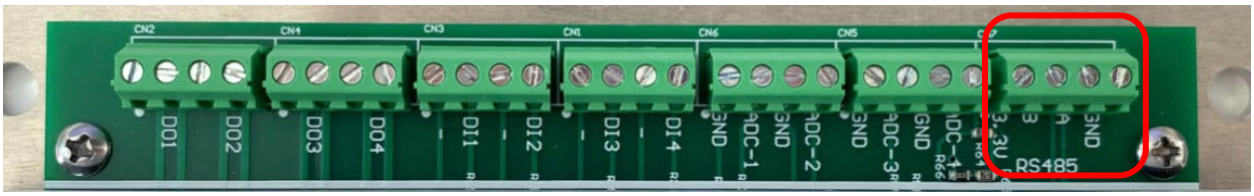
2.2.3 Analog Inputs:



Four 0 to 10 Volt analog inputs are available at ADC1, ADC2, ADC3, and ADC4. The board is labeled to show the negative poles of the sensors. The negative poles are connected to the negative poles of the battery and the positive to the sensor signal wire. Three analog sensors are available for use with the signal generation, such as a pressure sensor and temperature sensor that produces a 0 to 10 volts signal. ADC4 is set up to accommodate and PT100 temperature sensor. This provides cost-effective means to measure temperature.

Recommended to use ADC2 to connect with pressure transducer.

2.2.4 RS485



The final input block on the right side of the input bar is an RS485 connector. The left terminal of the block is a 3.3V output. The following terminal is the B terminal, then the A terminal, and finally, the final terminal is a ground terminal. This RS485 connection is a master terminal that, if programmed correctly, allows the controller logic in the ESP32 processor to control multiple motors through additional motor boards.

2.2.5 Side Access Connections:

The RS485 connectors shown in the picture below provide remote control access to the controller. The connection is a slave connector that provides remote monitoring and control of the motor functions from a SCADA system.



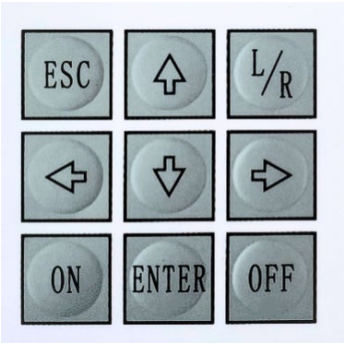
A third connection, a T-Tag connector, can be accessed through the lower opening on the side of the controller cover. This connection provides access to the lower motor board and is used to load the correct motor parameters for the motor and application.

2.3 Other features:

- A USB connection within the controller is provided for loading new programs in ESP32 chipset for different applications.
- The controller includes a real-time clock and an SD card. The real-time clock can be set form on the variables screen. The two can be programmed to provide a time-stamped permanent record of motor activity that can be accessed through the Bluetooth or WIFI capabilities of the ESP32 chipset.

3.0 Controller Operation – Compressors

3.1 Features Keypad Operations:



ESC

- Press to go back to the first menu page from another menu page
- When in a setting function, press to discard a setting and enter a new setting

L/R

- Cycle between local mode and remote control mode
  - set to local to give keyboard control
  - set to remote mode to accept remote control and communication.

WARNING

WHEN PERFORMING MAINTENANCE OR TESTING OF THE SYSTEM ONSITE. ENSURE THAT CONTROLLER IS SWITCHED TO "LOCAL" MODE, THEN SHUT DOWN THE SYSTEM AND ENSURE THAT CONTROLLER IS ISOLATED FROM THE POWER SOURCE. FAILURE TO DO SO MAY RESULT IN PERSONAL INJURY AND OR DAMAGE TO THE CONTROLLER. ONLY SWITCH THE SYSTEM BACK TO "REMOTE" MODE AFTER WORK ONSITE IS COMPLETE.

UP ARROW (↑) OR DOWN ARROW (↓)

- Press to scroll between the menu page
- In value setting mode, press once to increase or decrease a value by a fixed interval; press and hold to increase or decrease a value continuously

LEFT ARROW (←) OR RIGHT ARROW (→)

- Press to move to the item within a menu page;
- move the cursor to the left or right by one step when setting a value for a function

ENTER

- Press enter to activate the setting function for the item selected in a menu page
- after adjusting the settings, press enter to save the settings

ON – turns the system on
OFF – turns the system off. **Always turn off the system with the OFF button BEFORE disconnecting power to the motor and controller**

3.2 Menu Page Description:

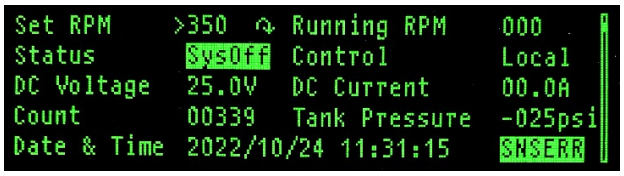
SCREEN 1 – 2: Home Page, Status, Setting

Connect the controller as described in the previous section. Close the switch to power up the controller. Shortly after powering up, the first screen will appear after a few seconds, which is also referred to as the Home Screen. To make changes on the screen, refer to section 3.1 to manipulate selector and cursor.

Screen 1



Screen 2

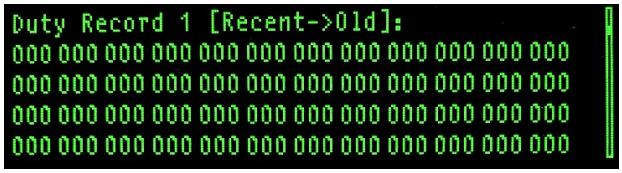


- "SET RPM": for motor speed. The speed range is 0 to 350 RPM. RPM setting is for starting speed. Motor will automatically change speed when experience higher load. ↷ indicates the motor will be turned in clockwise direction; ↶ indicates the motor will be turned in counter-clockwise direction.
- "RUNNING RPM": actual motor speed, may deviate from SET RPM due to load.
- "STATUS":
  - SYSON – the system is on and the motor state is to be confirmed
  - SYSOFF – the system is off and the motor state is off and the motor is off
  - MOn – the system is on and the motor is on
  - MOff –the system is on and the motor is off
  - SYSRUN – Normal running state
- "PRESSURE": pressure transducer reading. If reading is showing a negative value after connected with sensor with large tolerance, reference **Screen 6 item 5** for PSI offset
- "RUN TIME": cumulative run time of the motor; will only record when motor is turning; reset in each cycle time.
- "TIME ON": cumulative time of the controller being powered up despite motor status. Will reset upon each cycle time. Cycle time duration setting can be referenced in Compressor Setting **Screen 6 item 8**,
- "SET START PSI": pressure set point for compressor start up. This pressure has to be lower than SET STOP PSI by 3 PSI
- "SET STOP PSI": pressure set point for compressor shut down. This pressure has to be higher than SET START PSI by 3 PSI
- "DUTY %": compressor RUN TIME / TIME ON
- ERROR MSG: Right Conor of the screen indicates error message. Reference **Section 3.3** for details

SCREEN 3 : Duty Record

This screen displays the duty cycle records for 48 cycle time. Use ← or → to pull up Duty Record 2 page. Duty Record 1 page records instantaneous value, Duty Record 2 page stores records in memory. Reference Compressor Setting **Screen 6 item 8**, and **Screen 7 item 10** for cycle time setting and duty record setting. Return to Home Screen by pressing the ESC key.

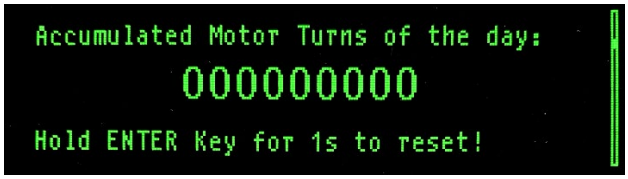
Screen 3



SCREEN 4 : Motor Turns Record

This screen displays the number of motor revolutions and can be reset either manually by user or automatically at midnight each day. Reference Compressor Setting **Screen 7 item 11** for reset selection. Return to Home Screen by pressing the ESC key.

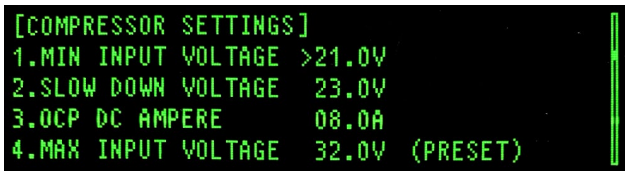
Screen 4



SCREEN 5 – 8 Compressor Settings

This screen allows the operator to set performance limits for the compressor.

Screen 5



Screen 6



Screen 7



Screen 8



- 1 “MIN INPUT VOLTAGE”: The minimum voltage to operate the motor, below which the system will shutdown
- 2 “SLOW DOWN VOLTAGE”: The power-saving voltage setting, below which the system will operate at ½ RPM to save power
- 3 “OCP DC AMP”: Motor will shut down if current is 1.5 times of setting for 15 seconds, warning if current is 1.2 times of setting for 2 minutes.
- 4 “MAX INPUT VOLTAGE”: Maximum voltage allowed to the motor. This value is has a default setting of 32 V, and cannot be changed.
- 5 “ZERO PSI OFFSET”: psi offset for altitude change. Reference table below

Initial Pressure (negative if higher than sea level)	PSI OFFSET VALUE
-4	1
-3	2
-2	3
-1	4
0 or positive	5 (default)

- 6 “SINGLE DUTY ALARM”: when the compressor duty over one cycle is higher than this value, DHIGH1 will be displayed on screen
- 7 “AVERAGE DUTY ALARM”: when the compressor duty averaged over 48 cycles is higher than this value, DHIGH2 will be displayed on screen.

SCREEN 5 – 8 Compressor Settings (CONTINUE)

Screen 5

```
[COMPRESSOR SETTINGS]
1.MIN INPUT VOLTAGE >21.0V
2.SLOW DOWN VOLTAGE  23.0V
3.OCP DC AMPERE      08.0A
4.MAX INPUT VOLTAGE  32.0V  (PRESET)
```

Screen 6

```
[COMPRESSOR SETTINGS]
5.ZERO PSI OFFSET     >05    [5 = 0 PSI]
6.SINGLE DUTY ALARM    100%
7.AVERAGE DUTY ALARM   100%
8.CYCLETIME PERIOD     00.5hour
```

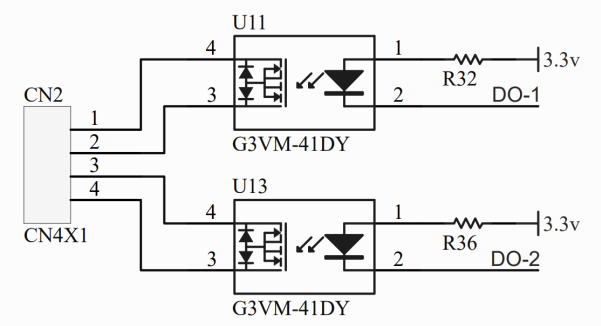
Screen 7

```
[COMPRESSOR SETTINGS]
9.PRESSURE CONTROL    >STOP MOTOR
10.CLEAR DUTY RECORDS  NO
11.MOTOR TURNS RESET   MANUAL
12.DISABLE DIO MODE    DISABLE
```

Screen 8

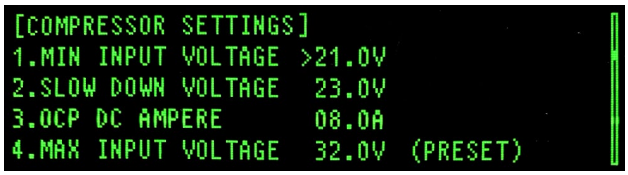
```
[COMPRESSOR SETTINGS]
13.COM FAIL ACTION    >NO ACTION
```

- 8 “CYCLETIME PERIOD”: this is the time duration for each cycle period to record motor RUN TIME and controller TIME ON. After each period, duty cycle will be calculated based on RUN TIME / TIME ON. valid settings are 0.5hr, 1hr, 1.5hr, 2hr, 3hr, 4hr, 6hr, 8hr, 12hr, 1day, 2days, 3days and 4days. Maximum 48 records can be stored in **Screen 3 duty record** memory.
- 9 “PRESSURE CONTROL”: controls motor setting when tank reaches Stop Pressure Setting value.
    - STOP MOTOR – Stop the motor when tank pressure reaches stop pressure
    - DOUT3 OPEN – The digital output number 3 on CN4 pin 1-2 will open and becomes high impedance
    - DOUT3 CLOSE - The digital output number 3 on CN4 pin 1-2 will close and becomes low impedance
    - If the PSI is out of control by 3psi higher than the stop pressure, the motor always stops despite what mode is selected.



SCREEN 5 – 8 Compressor Settings (CONTINUE)

Screen 5



Screen 6



Screen 7



Screen 8



- 10 “CLEAR DUTY RECORDS”: to clear and erase duty records memories in **Screen 3**
- 11 “MOTOR TURNS RESET”: to select manually or automatically reset recording number of motor revolutions in **Screen 4**
- 12 “DISABLE DIO MODE” : disable or enable DIO mode. Used for communication with local RTU or PLC.
- 13. “COME FAIL ACTION”: motor setting in the event of losing communication. No Action will ensure motor continue to run at the last setting prior to losing com. Stop Motor will shut down motor in the event of losing com or resume the last setting when com is recovered.

SCREEN 9 – 11: SCADA Setting

SCADA communications with the controller. The controller acts as a slave. The system can be accessed through connectors on the side of the controller as described in **Section 2.2.5**.

Screen 9

```
[SCADA MODBUS SETTINGS]
1.Protocol          >RTU     [RTU/ASCII]
2.Server ID          001     [1-247]
3.Timeout (ASCII)    02000   [100-65535]
4.Output Coil Addr   00000   [0-65533]
```

Screen 10

```
[SCADA MODBUS SETTINGS]
5.Discrete Input Addr >00000   [0-65535]
6.Input Register Addr  00000   [0-65535]
7.Hold Register Addr   00000   [0-65408]
```

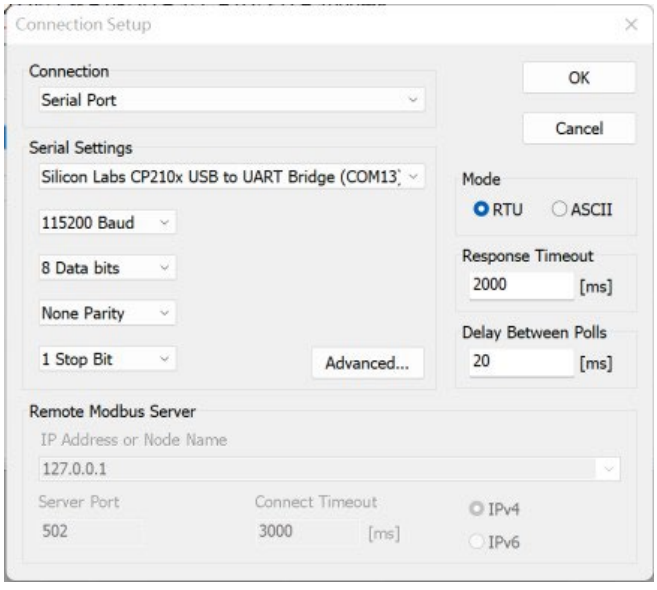
Screen 11

```
[SCADA SERIAL PORT SETTINGS]
1.Baud rate     >115200
2.Data bits      8
3.Stop bits      1
4.Parity         NONE
```

SCADA default setting:

Protocol	RTU
Server/Slave ID	1
Data bits	8
Parity	NONE
Stop bits	1
Baud Rates	115200bps

Example MODBUS & serial port setup:



SCREEN 9 – 11: SCADA Setting (CONTINUE)

SCADA communications with the controller. The controller acts as a slave. The system can be accessed through connectors on the side of the controller as described in **Section 2.2.5**.

Screen 9



Screen 10



Screen 11



Note:

- "Control" mode must be set to "Remote" in order to communicate with a SCADA system. Verify it if timeout is responded when polling the modbus.
- Changing SCADA/MODBUS settings requires a software reboot of the system to take effect; otherwise the SCADA/MODBUS will not work properly. You can press ESC when reboot is prompted, however, you must make sure to reboot the device by software or hardware when changing the last setting.
- Baud Rates of 9600bps to 115200bps are preferred. In general, faster is better for communication speed but poorer signal to longer RS485 wire length. Using 1200bps to poll too many registers at a time may interrupt the system, if this is the situation, use a higher baud rate or poll less registers at one time.

Reference Register Map **Section 3.4**

SCREEN 12: Fault History

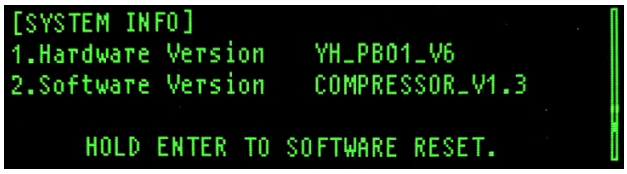
This screen lists the fault history and fault code for manufacturer's records. Return to Home Screen by pressing the ESC key. Press (←) or (→) to toggle between records. Hold "Enter" for 1 second to clear fault history, but NOT RECOMMENDED for troubleshooting.

Screen 12



SCREEN 13: Hardware and Software Information
This screen lists the hardware and software version of the controller and software for manufacturer's records. Return to Home Screen by pressing the ESC key.

Screen 13



SCREEN 14 Factory Setting
This screen is meant for manufacturer's factory records. Return to Home Screen by pressing the ESC key.

Screen 14



3.3 Error Message Description:

Error Message:
SETPAR – User is adjusting the system parameters
ERR001 – Invalid data range read from motor controller
ERR002 – HMI lost communication with motor controller
~~**ERR003 –** HMI cannot read back the motor state~~
PSIERR – Wrong PSI settings set by user
RPMERR - RPM set is too low
SNSERR – Pressure sensor error
OVPERR – DC input over-voltage error
UVPERR – DC input under-voltage error
BATLOW –DC input low-voltage warning
OCPWRN –Motor over current warning
OCPERR –Motor over current error
DHIGH1 –Single duty high alarm
DHIGH2–Average duty high alarm
REBOOT –Software reboot is pending to apply SCADA/COM settings
ESTERR – Internal controller error such as PID controller, Locked rotor etc.

3.4 Register Map:

Address	Coil	R/W	X	Unit	Type	Bit	Range	Description
BA1 + 0	**1**	R/W	/	/	UINT16	0-15	0x0000 – 0xFF00	Set system on/off state, 0xFF00/ON = on, 0x0000/OFF = off
BA1 + 1	**2**	R/W	/	/	UINT16	0-15	0x0000 – 0xFF00	Reset motor turns, 0xFF00/ON = reset, 0x0000/OFF = no action, read 0 means no reset is commanded or the last command was finished.
Address	**Register**	**R/W**	**X**	**Unit**	**Type**	**Bit**	**Range**	**Description**
BA2 + 0	1	R	/	/	UINT16	0	0 - 1	DC input over-voltage error, 1 = error, 0 = normal
						1	0 - 1	DC input under-voltage error, 1 = error, 0 = normal
						2	0 - 1	Motor over-current error, 1 = error, 0 = normal
						3	0 - 1	Wrong compressor PSI settings set by user, 1 = error, 0 = normal
						4	0 - 1	Invalid data range read from motor controller, 1 = error, 0 = normal
						5	0 - 1	Motor controller lost communication with HMI, 1 = error, 0 = normal
						6	0 - 1	EST controller error, 1 = error, 0 = normal
						7	0 - 1	Pressure sensor error, 1 = error, 0 = normal
						8	0 - 1	RPM set is too low, 1 = error, 0 = normal
						9-15	0 - 1	Read 0, reserved
BA2 + 1	2	R	/	/	UINT16	0	0 - 1	System on/off state, 1 = on, 0 = off
						1	0 - 1	Motor on/off state, 1 = on, 0 = off
						2	0 - 1	DC input low-voltage warning (motor slow down), 1 = warning, 0 = normal
						3	0 - 1	Motor over-current warning, 1 = warning, 0 = normal
						4	0 - 1	Single duty high warning, 1 = warning, 0 = normal
						5	0 - 1	Average duty high warning, 1 = warning, 0 = normal
						6	0 - 1	Reboot is pending, 1 = needed, 0 = no need
						7-15	0 - 1	Read 0, reserved
BA2 + 2	3	R/W	/	RPM	UINT16	0-15	0 - 350	Running RPM, write valid range = 20-350
BA2 + 3	4	R	10	V	UINT16	0-15	0 - 430	DC input voltage, 430 = 43.0V
BA2 + 4	5	R	10	A	UINT16	0-15	0 - 165	DC input current, 165 = 16.5A
BA2 + 5	6	R	10	A	UINT16	0-15	0 - 165	Motor coil current (RMS) , 165 = 16.5A
BA2 + 6	7	R/W	/	PSI	UINT16	0-15	1 - 87	Start pressure
BA2 + 7	8	R/W	/	PSI	UINT16	0-15	4 - 90	Stop pressure
BA2 + 8	9	R	/	PSI	UINT16	0-15	0 - 100	Tank pressure
BA2 + 9	10	R/W	/	yyyy	UINT16	0-15	2000 - 2099	Date: Year, write valid range = 2022 - 2099
BA2 + 10	11	R/W	/	mm	UINT16	0-15	1 - 12	Date: Month, if day is larger than maximum day of the month, day will be changed automatically
BA2 + 11	12	R/W	/	dd	UINT16	0-15	1 - 31	Date: Day, write valid range depends on month
BA2 + 12	13	R/W	/	hh	UINT16	0-15	0 - 23	Time: Hour
BA2 + 13	14	R/W	/	mm	UINT16	0-15	0 - 59	Time: Minute
BA2 + 14	15	R/W	/	ss	UINT16	0-15	0 - 59	Time: Second
BA2 + 15	16	R	/	day(s)	UINT16	0-15	0 - 65535	Compressor time on: day, maximum value is equivalent to 179 years
BA2 + 16	17	R	/	hh	UINT16	0-15	0 – 23	Compressor time on: hour
BA2 + 17	18	R	/	mm	UINT16	0-15	0 – 59	Compressor time on: minute
BA2 + 18	19	R	/	ss	UINT16	0-15	0 – 59	Compressor time on: second
BA2 + 19	20	R	/	day(s)	UINT16	0-15	0 – 65535	Compressor run time: day, maximum value is equivalent to 179 years
BA2 + 20	21	R	/	hh	UINT16	0-15	0 – 23	Compressor run time: hour
BA2 + 21	22	R	/	mm	UINT16	0-15	0 – 59	Compressor run time: minute
BA2 + 22	23	R	/	ss	UINT16	0-15	0 – 59	Compressor run time: second
BA2 + 23	24	R	/	%	UINT16	0-15	0 – 100%	Compressor duty
BA2 + 24	25	R/W	/	%	UINT16	0-15	0 – 100%	Single duty alarm
BA2 + 25	26	R/W	/	%	UINT16	0-15	0 – 100%	Average duty alarm
BA2 + 26	27	R/W	10	hour	UINT16	0-15	0 – 960	Compressor cycle time set, write valid values are:5(0.5hr),10(1hr),15(1.5hr),20(2hr),30(3hr),40(4hr),60(6hr),80(8hr),120(12hr),240(1day),480(2days),720(3days),960(4days)
BA2 + 27	28	R/W	/	/	UINT16	0-15	0 – 2	Pressure control method set, 0 – Stop motor when pressure is reached 1 – DOUT1 opens and becomes HIGH Z 2 – DOUT1 closes and becomes LOW Z
BA2 + 28	29	R	/	/	UINT16	0-15	0 – 65535	Motor turns (UINT32)
BA2 + 29	30	R	/	/	UINT16	0-15	0 – 65535	= (Register 29 << 16 | Register 30)
BA2 + (30-77)	31-78	R	/	%	UINT16	0-15	0 – 100	Duty record 1(volatile): Data 0 – 47, new to old order averaged by compressor cycle time set
BA2 + (78-125)	79-126	R	/	%	UINT16	0-15	0 – 100	Duty record 2(non-volatile): Data 0 – 47, new to old order averaged by duty record 1
BA2 + 126	127	R/W	/	/	UINT16	0-15	0 – 1	System on/off state, 1 = on, 0 = off

*Programmable Output Coil Base Address (BA1) is set in HMI

*Programmable Holding Register Base Address (BA2) is set in HMI

Warning: Do NOT perform non-stop write action; change will be saved every time to the non-volatile memory, which has limited write cycles

4.0 Controller Back Mounting Plate Drawing:

